Code No: 135AF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
DESIGN AND ANALYSIS OF ALGORITHMS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

USED PAPERS SEP-2021

1. Design an algorithm to sort the given list of elements using Quick Sort incorporating divide and conquer technique. Sort the following list using the same and compute its best case time efficiency: 4, 2, 0, 8, 7, 1, 3, 6. [15]

2. Consider the following recurrence
   $T(n)=3T(n/3)+ n$
   Obtain asymptotic bound using substitution method. [15]

3. Using disjoint-sets find the connected components in the undirected graph G = (V, E), where the vertices V = {a, b, c, d, e, f, g, h, i, j} and edges E = {(a, c), (a ,b), (e, f), (h, i), (e, g), (a, d), (e, d) (b, d), (c, a), (h, j)}. The edges are processed in the order given. List the vertices in each connected component after each step. [15]

4. Find a solution to the 8-Queens problem using backtracking strategy. Draw the solution space using necessary bounding function. [15]

5. Suppose we run Dijkstra's single source shortest-path algorithm on the following edge weighted directed graph with vertex P as the source (figure 1). In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized? [15]

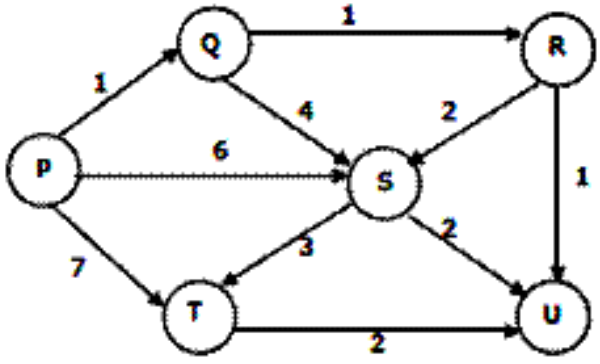


Figure 1

6. A thief enters a house for robbing it. He can carry a maximal weight of 60 kg into his bag. There are 5 items in the house with the following weights and values. What items should thief take if he can even take the fraction of any item with him? (w1, w2, w3, w4, w5) = (5, 10, 15, 22, 25), (b1, b2, b3, b4, b5) = (30, 40, 45, 77, 90). [15]

7. Given a set of cities and distance between every pair of cities, the problem is to find the shortest possible route that visits every city exactly once and returns to the starting point. (figure 2) [15]

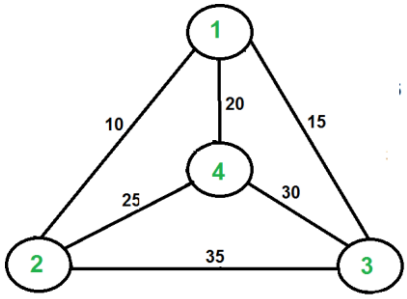


Figure 2

8. Given a set of 4 items, each with weight {2,2,4,5} and benefit {3,7,2,9}, Using LC branch and Bound determine the items to include in the knapsack so that the total weight is less than or equal to a given weight limit and the total benefit is maximized. The weight limit for this knapsack is 10? [15]

---ooOoo---

USED PAPERSSEP-2021